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EXP3:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main() {
    char key[] = "MONARCHY", s[100], m[5][5];
    int used[26] = {0}, i, j, k = 0;
    int r1, c1, r2, c2, n;
    for (i = 0; key[i]; i++) {
        char ch = toupper(key[i]);
        if (ch == 'J') ch = 'I';
        if (!used[ch - 'A']) {
            m[k / 5][k % 5] = ch;
            used[ch - 'A'] = 1;
            k++;
        }
    }
    for (i = 0; i < 26; i++)
        if (i != 9 && !used[i]) {
            m[k / 5][k % 5] = 'A' + i;
            k++;
        }
    printf("Enter plaintext (even length): ");
    scanf("%99s", s);
    n = strlen(s);

    for (i = 0; i < n; i++)
        s[i] = toupper(s[i]) == 'J' ? 'I' : toupper(s[i]);
```

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    for (i = 0; i < n; i += 2) {
        r1 = r2 = c1 = c2 = -1;
        for (j = 0; j < 25; j++) {
            if (m[j / 5][j % 5] == s[i])
                r1 = j / 5, c1 = j % 5;
            if (m[j / 5][j % 5] == s[i + 1])
                r2 = j / 5, c2 = j % 5;
        }
        if (r1 == r2) {
            s[i] = m[r1][(c1 + 1) % 5];
            s[i + 1] = m[r2][(c2 + 1) % 5];
        } else if (c1 == c2) {
            s[i] = m[(r1 + 1) % 5][c1];
            s[i + 1] = m[(r2 + 1) % 5][c2];
        } else {
            s[i] = m[r1][c2];
            s[i + 1] = m[r2][c1];
        }
    }
    printf("Ciphertext: %s", s);
    return 0;
}
```

OUTPUT:

